

99.500 Applied Mathematics for Engineering

Sample Lab Test Question

April 9, 2026

2. Solve the following dimensionless PDE by using “pdepe” in Matlab when $\delta = 1, 5$ and 10 .

$$\frac{\partial T}{\partial t} = \frac{1}{x} \frac{\partial}{\partial x} \left(x \frac{\partial T}{\partial x} \right) \quad (1)$$

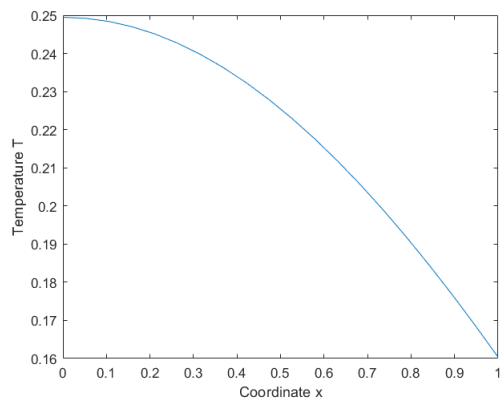
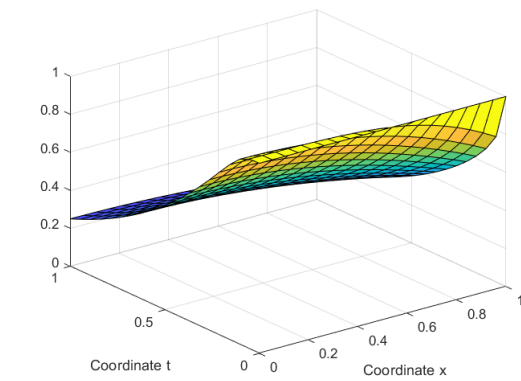
$$T(x, 0) = 1 \quad (2)$$

$$\left. \frac{\partial T}{\partial x} \right|_{x=0} = 0 \quad (3)$$

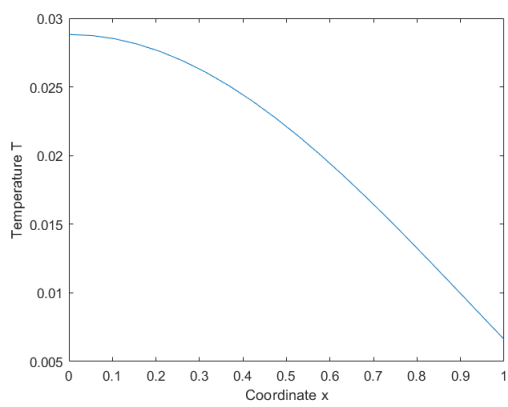
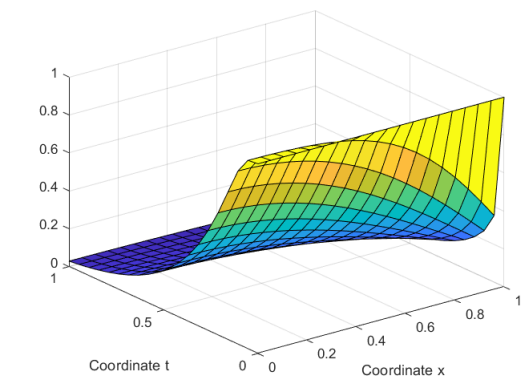
$$\left. \frac{\partial T}{\partial x} \right|_{x=1} + \delta \cdot T(1, t) = 0 \quad (4)$$

where $T = T(x, t)$ and δ is a constant. Show the results in 3 “surface” graphs when $0 \leq t \leq 1$, as well as the plots of T vs x when $t = 1$.

Solution: $\delta = 1$



Solution: $\delta = 5$



Solution: $\delta = 10$

